

MTH208 : Data Science Lab I

- **Course Description:** The course equips students with a fundamental computational learning base for modern data analysis. Focus will be first on logical coding skills, and simulating experiments. Further we will focus on collecting, cleaning, and organizing data, and presenting clean insights via interactive web-based apps. We will also study code-benchmarking, and how to integrate C++ codes in R. Programming languages used will be mainly R.

- **Prerequisites:** None.

- **Schedule:**

Location:: Room 204H, Diamond Jubilee Building

Timings:: T: 9:00am - 10:30am, Th: 2:00pm - 3:30pm

- **Structure:** This is a lab course with 2 lab classes, each of length 1.5 hours. The style of the lab is worksheet based, so in most labs there will be a worksheet given to you that you have to work through. The goal is to complete the whole worksheet in the 1.5 hours of the lab.

The worksheet will be mostly self-sufficient, but often we will explain solutions, concepts, and key steps. The goal is to not take anything home to do.

- **Course Team:**

– **Instructor:**

* Akash Anand

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– **Teaching Assistant:**

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- **Course Page:**

The Mookit (HelloIITK) course page is available at <https://hello.iitk.ac.in>. You should be able to sign in using your CC login credentials to access the course webpage.

- **Course Grading Policy:**

The following are the evaluation components with respective credit distribution

Lab Quizzes	20%
Project	20%
Mid-sem Exam	30%
Final Exam	30%

The final grade will be relative.

- **Project:**

A significant component of the course will be a group project. Below will be the following components of the Project:

- (1) Obtain data by scraping
- (2) Ask questions about the data
- (3) Do exploratory data analysis
- (4) Make a Shiny app on the data analysis
- (5) Make a presentation
- (6) Write a report

- **Course Policies:**

The course should be cleared via honest practices and any unfair practice will be dealt with seriously. No pro rata of marks will be done for missing exams for non-medical reasons. However, emergency medical situations will be dealt on a case to case basis subject to approval and verification by DUGC/DPGC and SUGC/SPGC.

- **References:**

- (1) Student-sourced book to prepare for this course:
<https://dvats.github.io/BasicRProblems/>
- (2) Wickham, Hadley. “Elegant graphics for data analysis.” O’Reilly Media 35.211 (2009)
- (3) Wickham, Hadley. “Mastering shiny”. O’Reilly Media, Inc., 2021.
- (4) Bruce, Peter, Andrew Bruce, and Peter Gedeck. “Practical statistics for data scientists: 50+ essential concepts using R and Python”. O’Reilly Media, 2020.
- (5) VanderPlas, Jake. “Python data science handbook: Essential tools for working with data.” O’Reilly Media, Inc., 2016.
- (6) Boehmke, Bradley C. “Data wrangling with R”. Springer International Publishing, 2016.
- (7) Pineau, Joelle, et al. “Improving reproducibility in machine learning research (a report from the Neurips 2019 reproducibility program).” Journal of Machine Learning Research 22 (2021).